THEORY LABORATORY FOR MODERN ROBOTICS

Training Set 4: Tensors on manifolds

Online discussion of model solutions (sets 3 and 4) on Monday, 8 June 2020, 14:00–16:00

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1. **Theoretical developments**

   Let $(M, \mathcal{O}, \mathcal{A})$ be a smooth manifold.

   (a) Show that the set $C^\infty(M) := \{f : M \to \mathbb{R} \text{ smooth}\}$ equipped with a pointwise defined addition $\boxplus$ and S-multiplication $\boxdot$ constitutes a vector space $(C^\infty(M), \boxplus, \boxdot)$. Can you find an infinite set E of elements in $C^\infty(M)$ such that every finite subset $F \subset E$ is linearly independent? What does this imply for the dimension of the vector space?

   (b) For a smooth manifold $(M, \mathcal{O}, \mathcal{A})$, the set $T_pM := \{X_{\gamma,p} \,|\, \gamma : \mathbb{R} \to M \text{ with } \gamma(0) = p\}$ of tangent vectors at a point $p \in M$ can be equipped with a map $\oplus : T_pM \times T_pM \to T_pM$ given by letting

   $$X_{\gamma,p} \oplus X_{\delta,p} := X_{\alpha,p}\,,$$

   where the curve α on M is defined in terms of the curves γ and δ on M as

   $$\alpha(\lambda) = x^{-1}((x \circ \gamma)(\lambda) + (x \circ \delta)(\lambda) - x(p))$$

   with respect to any chart $(U, x) \in \mathcal{A}$ with $p \in U$. Prove that choosing any other chart $(V, y) \in \mathcal{A}$ with $p \in V$ yields the same tangent vector $X_{\gamma,p}$ and thus the same map $\oplus$. Why is this important?

   (c) Let X be an element of the tangent bundle TM. Explain why choosing a chart $(U, x) \in \mathcal{A}$ with $\pi(X) \in U$ and another chart (V, y) with $\pi(X) \in V$ allows to write, alternatively,

   $$X = \sum_{a=1}^{\dim M} X^a_{(U,x)} \left(\frac{\partial}{\partial x^a}\right)_{\pi(X)} \qquad \text{or} \qquad X = \sum_{a=1}^{\dim M} X^a_{(V,y)} \left(\frac{\partial}{\partial y^a}\right)_{\pi(X)}$$

   for unique real numbers $X^1_{(U,x)}, \ldots, X^{\dim M}_{(U,x)}$ and $X^1_{(V,y)}, \ldots, X^{\dim M}_{(V,y)}$, respectively. Show from these two alternative expansions of X in terms of the two chart-induced bases that the respective components are related as

   $$X^a_{(U,x)} = \sum_{b=1}^{\dim M} \left(\frac{\partial x^a}{\partial y^b}\right)_{\pi(X)} X^b_{(V,y)}\,.$$

   (d) Let $\mathcal{X} : M \to TM$ and $\mathcal{Y} : M \to TM$ be two vector fields on M and $f \in C^\infty(M)$. First show that

   $$\mathcal{X}f : M \to \mathbb{R}\,, \qquad (\mathcal{X}f)(p) := \mathcal{X}(p)f$$

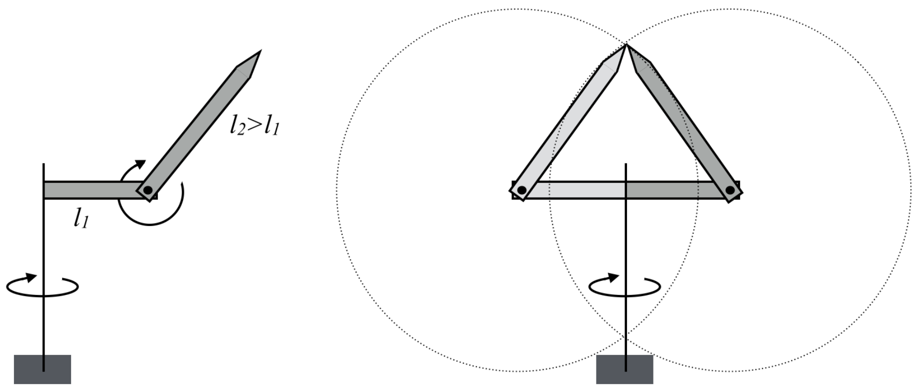
   is a smooth function on M. Now consider the so-called *Lie bracket* $[\mathcal{X}, \mathcal{Y}]$ defined by its action of an arbitrary function $f \in C^\infty(M)$ through

   $$[\mathcal{X}, \mathcal{Y}]f := \mathcal{X}(\mathcal{Y}f) - \mathcal{Y}(\mathcal{X}f)\,.$$

   Obviously, $[\mathcal{X}, \mathcal{Y}]f$ is a smooth function on M since both $\mathcal{X}(\mathcal{Y}f)$ and $\mathcal{Y}(\mathcal{X}f)$ are. However, show that while neither $\mathcal{X}\mathcal{Y}$ nor $\mathcal{Y}\mathcal{X}$ are vector fields, the Lie bracket $[\mathcal{X}, \mathcal{Y}]$ is a vector field on M (Hint: Evaluate $\mathcal{X}(\mathcal{Y})f$ and $\mathcal{Y}(\mathcal{X})f$ at some point p in the domain U of some chart $(U, x) \in \mathcal{A}$ and explain why only their difference, but none of them individually, presents a directional derivative of f at p).

2. **Practical application: A simple robot whose description *requires* a manifold**

Consider a robot with a short arm of length l_1 mounted on a vertical axis and a longer arm of length $l_2 > l_1$ mounted on a perpendicular axis at the end of the short arm, see figure (a).



(a) (b)

(a) Choosing the mounting point of the short arm on the vertical axis as the origin, define the set $P \subset \mathbb{R}^3$ of all possible points that can be reached by the tip of the long arm, if the distance of any two points in $\mathbb{R}^3$ is given by

$$\|(a,b,c) - (d,e,f)\| := \sqrt{(a-d)^2 + (b-e)^2 + (c-f)^2}\,.$$

(b) Considering figure (b), explain why P *fails* to represent all possible the configurations of the robot.

(c) Define the set $C_1 := \{(x,y) \in \mathbb{R}^2 \,|\, x^2 + y^2 = l_1{}^2\}$ and consider it equipped with the subset topology $\mathcal{O}|_{\mathbb{R}^2}$ defined in problem 2(a) of training set 3. Provide two charts (U,x) and (V,y) that cover C_1 (which simply means that $C_1 = U \cup V$) and reveal C_1 as a one-dimensional topological manifold.

(d) Check whether the two charts you constructed in (b) constitute a smooth atlas. If not, replace one of the two charts by another one such that the atlas becomes smooth.

(e) Now consider the set $C_1 \times C_2$, where C_1 is as above and $C_2 := \{(z,w) \in \mathbb{R}^2 \,|\, z^2 + w^2 = l_2{}^2\}$. Consider the set $C_1 \times C_2$ equipped with the product topology $\mathcal{O}_{C_1 \times C_2}$ defined in problem 2(b) of training set 3. Use the two charts for C_1 from part (c) and two analogously constructed charts for C_2 in order to construct *four* charts that constitute a smooth atlas $\mathcal{A}_{C_1 \times C_2}$ for the topological manifold $(C_1 \times C_2, \mathcal{O}_{C_1 \times C_2})$.

(f) What is the dimension of the thus defined smooth manifold $(C_1 \times C_2, \mathcal{O}_{C_1 \times C_2}, \mathcal{A}_{C_1 \times C_2})$? Explain why $C_1 \times C_2$ represents the configuration space of the robot and how it differs from the set P defined above.

(g) Provide a smooth curve γ in $C_1 \times C_2$ that connects the two different configurations of the robot shown in figure (b) such that the tip of the long arm never moves. Identify all charts through whose domains this curve runs and provide the respective chart representations of the curve.

(h) Calculate the tangent vectors to the curve γ at the two points corresponding to the two configurations shown in figure (b).

(i) Do you see how badly needed smooth manifolds are in order to describe the configuration space (and much more!) for even the simplest robots?